

Answer **all** the questions.

- 1 Calculate $\frac{1458}{23.4 - 3.699^2}$. Write your answer correct to

(a) 4 decimal places,

Answer [1]

(b) 4 significant figures.

Answer [1]

- 2 A water tank contains 2.47×10^7 drops of water and 7×10^9 micro-organisms.
Let L be the average number of micro-organisms per drop of water.
Find the value of L . Give your answer in standard form.

Answer [2]

- 3 (a) Of the 195 countries in the world, Ryan has visited 20. What percentage of the countries in the world has Ryan visited?

Answer % [1]

- (b) At a sale, all prices are reduced by 22.5%. The price of a set of waterproof headphones during the sale is \$139.50. Find its original price.

Answer \$ [1]

- 4 Show that $3^{3x+2} - 9^{\frac{3}{2}x} + (27)^{x+1}$ is divisible by 5 for all positive integer values.

Answer

[2]

- 5 (a) The highest common factor and lowest common multiple of three numbers are 12 and 2376 respectively. If two of the numbers are 108 and 72, find the smallest possible integer value of the third number.

Answer [2]

- (b) (i) Express 6468 as a product of its prime factors.

Answer [1]

- (ii) Find the smallest positive integer k , such that $\sqrt{6468k}$ is an integer.

Answer $k =$ [1]